

Customer Shopping Behaviour Analysis

1. Project Overview

This project analyzes customer shopping behaviour using transactional data from 3,900 customer purchases across various product categories. The primary objective is to uncover insights into spending patterns, customer demographics, product preferences, and subscription behaviour to support data-driven business decisions.

Project Objectives

- Analyze customer purchasing behaviour.
- Identify revenue-generating customer segments.
- Study subscription patterns and customer loyalty.
- Understand category-wise revenue and sales distribution.
- Build an interactive Power BI dashboard for business insights.
- Provide actionable business recommendations.

2. Dataset Summary

Dataset Information

- Total Records: 3,900
- Total Columns: 18

Key Features

Customer Information

- Customer ID
- Age
- Gender
- Location

Product Information

- Item Purchased
- Category
- Color
- Size
- Season

Transaction Information

- Purchase Amount
- Previous Purchases
- Payment Method
- Shipping Type
- Subscription Status
- Discount Applied
- Review Rating

Engineered Features

- Age Group
- Purchase Frequency Days

Data Quality

- Dataset contains 3,900 customer purchase records.
- Missing values were checked and handled appropriately.
- Column names were standardized for analysis.

[12]:

```
import pandas as pd
df = pd.read_csv(
    r"C:\Users\om\OneDrive\Desktop\customer_behaviour_analysis\customer_shopping_behavior.csv"
)
df.head()
```

[12]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

3. Data Preparation and Feature Engineering

The dataset was cleaned, transformed, and prepared using Python and Pandas before loading it into MySQL and Power BI.

Data Cleaning Steps

- Converted column names to lowercase.
- Replaced spaces with underscores.
- Checked for missing values.

- Verified data types.
- Removed unnecessary columns.

Feature Engineering

Age Group Creation

Customers were segmented into:

- Young Adult
- Adult
- Middle-aged
- Senior

Purchase Frequency Days

A new numerical feature was created to represent customer purchase frequency for analysis and visualization purposes.

Technologies Used

- Python
- Pandas
- Jupyter Notebook

```
[15]: df.isnull().sum()
```

```
[15]: Customer ID      0
      Age              0
      Gender           0
      Item Purchased   0
      Category         0
      Purchase Amount (USD)  0
      Location         0
      Size             0
      Color            0
      Season           0
      Review Rating    37
      Subscription Status  0
      Shipping Type    0
      Discount Applied  0
      Promo Code Used  0
      Previous Purchases  0
      Payment Method   0
      Frequency of Purchases  0
      dtype: int64
```

```
16]: df['Review Rating'] = df.groupby('Category')['Review Rating'] \
      .transform(lambda x: x.fillna(x.median()))
```

```
17]: df.isnull().sum()
```

```
17]: Customer ID      0
      Age             0
      Gender          0
      Item Purchased  0
      Category        0
      Purchase Amount (USD) 0
      Location        0
      Size            0
      Color           0
      Season          0
      Review Rating   0
      Subscription Status 0
      Shipping Type   0
      Discount Applied 0
      Promo Code Used 0
      Previous Purchases 0
      Payment Method  0
      Frequency of Purchases 0
      dtype: int64
```

Missing values were identified and handled during preprocessing.

```
[21]: # create a column age_group

      labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']

      df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

```
[22]: df[['age', 'age_group']].head(10)
```

```
[22]:
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

4. SQL Business Analysis

After data preparation, the dataset was loaded into MySQL for business analysis.

```
[29]: from sqlalchemy import create_engine

      username = "root"
      password = "password"
      host = "localhost"
      port = "3306"
      database = "customer_behavior"

      engine = create_engine(
          f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
      )

[30]: table_name = "customer"

      df.to_sql(
          table_name,
          engine,
          if_exists="replace",
          index=False
      )

      print(f"Data successfully loaded into table '{table_name}'")

      Data successfully loaded into table 'customer'
```

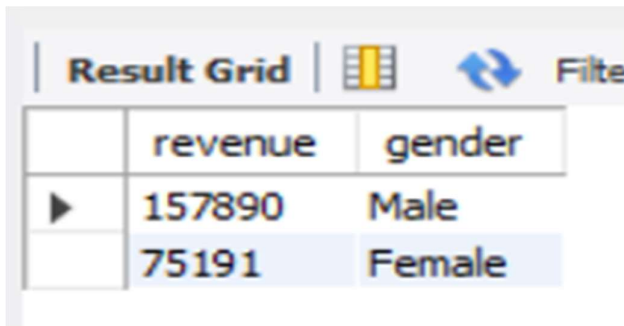
The following business questions were answered using SQL queries.

1. Revenue by Gender

Determine the total revenue generated by male and female customers.

Findings

- Male customers generated higher revenue than female customers.
- Gender-based purchasing patterns can help target marketing campaigns.



The screenshot shows a Jupyter Notebook interface with a 'Result Grid' tab. It displays a table with two columns: 'revenue' and 'gender'. The first row shows a revenue of 157890 for Male customers, and the second row shows a revenue of 75191 for Female customers. A play button icon is visible next to the first row.

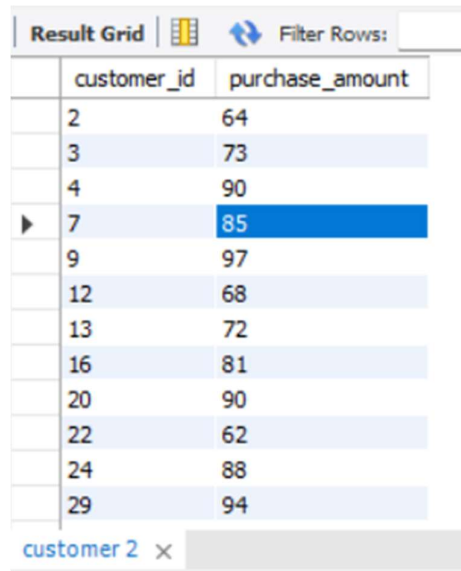
	revenue	gender
▶	157890	Male
	75191	Female

2. High-Spending Discount Users

Identify customers who used discounts but still spent more than the average purchase amount.

Findings

- Several customers continued spending above average despite receiving discounts.
- These customers represent valuable target segments.



	customer_id	purchase_amount
	2	64
	3	73
	4	90
▶	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94

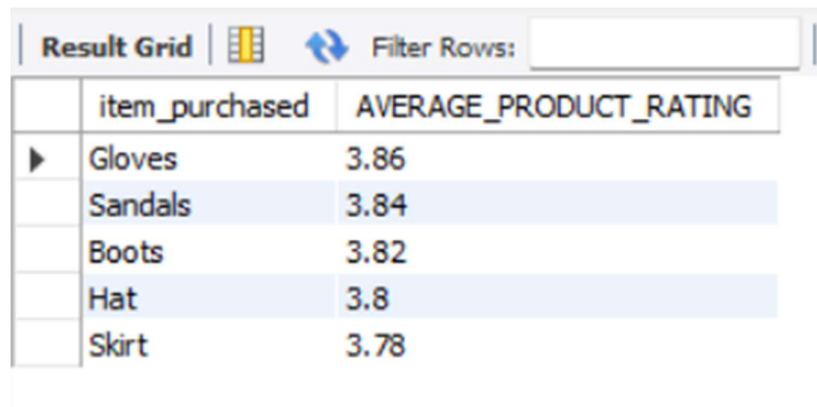
customer 2 ×

3. Top 5 Products with Highest Average Review Ratings

Determine products receiving the highest customer satisfaction ratings.

Findings

- Highly rated products indicate strong customer satisfaction.
- These products should be prioritized in promotional campaigns.



	item_purchased	AVERAGE_PRODUCT_RATING
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. Shipping Type Analysis

Compare average purchase amounts across shipping methods.

Findings

- Different shipping options influence purchasing behaviour.
- Premium shipping customers often demonstrate higher purchase values.

Result Grid			Filter Rows:	
	shipping_type	ROUND(AVG(purchase_amount),2)		
▶	Express	60.48		
	Standard	58.46		

5. Subscriber Spending Analysis

Compare spending behaviour between subscribers and non-subscribers.

Findings

- Subscribers and non-subscribers exhibit different spending patterns.
- Subscription programs can influence customer retention and revenue generation.

Result Grid					Filter Rows:		Export:	Wrap
	total_customers	subscription_status	avg_spend	total_revenue				
▶	1053	Yes	59.49	62645				
	2847	No	59.87	170436				

6. Discount Effectiveness Analysis

Identify products with the highest percentage of purchases involving discounts.

Findings

- Certain products rely heavily on discount-based sales.
- Useful for pricing and promotional strategies.

Result Grid			Filter Rows:	
	item_purchased	discount_rate		
▶	Hat	50.00		
	Sneakers	49.66		
	Coat	49.07		
	Sweater	48.17		
	Pants	47.37		

7. Customer Segmentation

Customers were classified into three categories based on previous purchases:

- New Customers
- Returning Customers
- Loyal Customers

Findings

- Returning customers form a significant portion of the customer base.
- Loyal customers contribute to long-term business growth.

Result Grid		Filter Rows:	
	customer_segment	Number of Customers	
▶	Loyal	3116	
	Returning	701	
	New	83	

8. Top 3 Products per Category

Identify the most purchased products within each product category.

Findings

- Reveals customer preferences within each category.
- Supports inventory and product planning decisions.

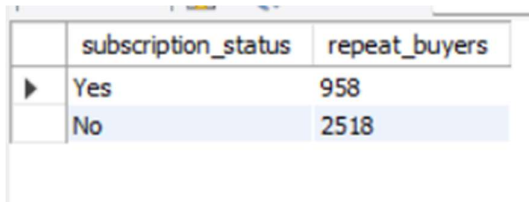
Result Grid		Filter Rows:		Export:	
	item_rank	category	item_purchased	total_orders	
▶	1	Accessories	Jewelry	171	
	2	Accessories	Sunglasses	161	
	3	Accessories	Belt	161	
	1	Clothing	Blouse	171	
	2	Clothing	Pants	171	
	3	Clothing	Shirt	169	
	1	Footwear	Sandals	160	
	2	Footwear	Shoes	150	
	3	Footwear	Sneakers	145	
	1	Outerwear	Jacket	163	
	2	Outerwear	Coat	161	

9. Repeat Buyer Analysis

Analyze whether customers with more than five previous purchases are also likely to subscribe.

Findings

- Repeat buyers show stronger engagement with the business.
- Subscription adoption among repeat customers can improve retention.



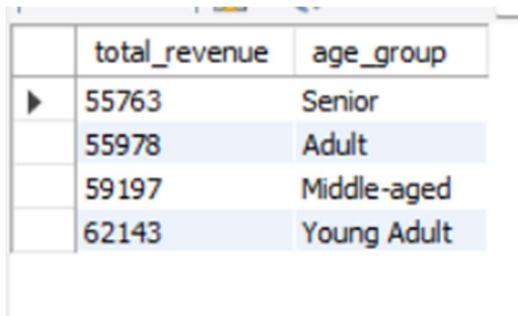
	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

10. Revenue Contribution by Age Group

Calculate total revenue generated by each age group.

Findings

- Different age groups contribute differently to overall revenue.
- Helps target customer segments more effectively.



	total_revenue	age_group
▶	55763	Senior
	55978	Adult
	59197	Middle-aged
	62143	Young Adult

5. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize customer behaviour and business performance.

Dashboard KPIs

- Number of Customers
- Total Revenue
- Average Purchase Amount
- Average Review Rating

Dashboard Filters

- Subscription Status

- Gender
- Category
- Shipping Type

Dashboard Visualizations

Revenue Analysis

- Revenue by Category
- Revenue by Age Group

Sales Analysis

- Sales by Category
- Sales by Age Group

Customer Analysis

- Customer Distribution by Subscription Status

Dashboard Features

- Interactive slicers
- Dynamic KPI updates
- Category-level filtering
- Shipping-type filtering
- Clear All Slicers functionality



6. Key Insights

Customer Base

- Total Customers: 3,900

Revenue Performance

- Total Revenue: \$233.08K
- Average Purchase Amount: \$59.76

Customer Satisfaction

- Average Review Rating: 3.75

Subscription Analysis

- Approximately 73% customers are non-subscribers.
- Approximately 27% customers are subscribers.

Category Performance

- Clothing generates the highest revenue.
- Clothing records the highest sales volume.

Age Group Performance

- Young Adults contribute the highest revenue.
- Young Adults also generate the highest sales volume.

7. Business Recommendations

Increase Subscription Adoption

Provide exclusive offers, discounts, and loyalty rewards to encourage more customers to subscribe.

Focus on High-Performing Categories

Prioritize Clothing and Accessories categories through targeted marketing campaigns and inventory optimization.

Personalized Marketing

Create customer-specific campaigns based on age groups, purchasing patterns, and subscription status.

Customer Retention Strategies

Develop loyalty programs targeting repeat and high-value customers.

Optimize Shipping Services

Analyze customer preferences for shipping methods and improve logistics efficiency.

Promote Highly Rated Products

Leverage top-rated products in promotional campaigns to increase customer confidence and sales.

8. Conclusion

This project successfully demonstrates an end-to-end data analytics workflow using Python, MySQL, SQL, and Power BI.

The analysis identified customer purchasing trends, spending behaviour, subscription patterns, and category performance. An interactive dashboard was developed to provide business stakeholders with meaningful insights and support strategic decision-making.

The project highlights practical applications of data cleaning, feature engineering, SQL analysis, dashboard development, and business intelligence reporting in a real-world customer analytics scenario.

Tools and Technologies Used

- Python
- Pandas
- Jupyter Notebook
- MySQL
- SQL
- Power BI
- GitHub